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Revision History

Name	Date	Reason For Changes	Version
Connie Rodriguez	10/20/24	Updated Section 1	1.1
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Connie Rodriguez	11/4/24	Updated Section 3	1.3
Connie Rodriguez	11/25/24	Updated Section 4 and Appendix	1.4

1. Introduction

1.1 Purpose

The purpose of this document is to provide a comprehensive description of the Washington Vets 2 Tech (WAV2T) platform, an educational system designed to support prospective, current, and graduated students in their academic and professional journeys. This system provides tools for skills assessment, adaptive practice exams, study materials, and entrance exams tailored to three different learning pathways known as cohorts. Additionally, it integrates a chatbot for study assistance and resources for professional certification preparation.

This document details the system's purpose, features, interfaces, functionality, and constraints. It also explains how the system will interact with external inputs and environments. The document serves as a reference for the WAV2T development team, Dr. Dvorak, and the Engineering Advisory Board (EAB), ensuring alignment with the project's objectives and securing approval for further development.

1.2 Scope of Project

The WAV2T platform is an educational system that empowers students at all stages of their journey—prospective, current, and graduated—by offering tools and resources designed to foster success. The platform includes the following features:

- Secure Login System: Protects user data and provides access to tailored resources.
- Entrance Exams: Customized for each of the three pathways to determine readiness and placement.
- Skills Assessment: Evaluates user abilities to guide their learning path.
- Adaptive Practice Exams: Dynamically adjusts to the user's skill level for more effective learning.
- Study Materials: Comprehensive resources for entering the program, succeeding within it, and achieving professional certifications.
- Chatbot Study Buddy: An AI-powered assistant to support users with studying and resource navigation.
- Math Evaluation Exam: Ensures students meet pre-requisite requirements for program success.

- **Data Analysis for Admins:** Provides insights for the WAV2T team to assess program impact and secure ongoing funding.
- **Multi-Tabbed Layout:** Organized interface guiding users through the various educational tools and resources.

The WAV2T platform aims to:

1. **Streamline the Onboarding Process:** Provide prospective students with the tools to succeed in entrance exams and math prerequisites.
2. **Enhance Student Success:** Offer adaptive study materials and assessments for current students.
3. **Support Graduated Students:** Deliver resources to help alumni achieve industry-recognized certifications.
4. **Enable Data-Driven Decision-Making:** Equip administrators with analytics for continuous program improvement.

1.2.1 Key Technologies

The development of the WAV2T platform will leverage several key technologies to ensure a secure, and user-friendly system:

- **ASP.NET Core Framework:** Provides a secure and scalable foundation for backend development, enabling efficient handling of user authentication, database operations, and application programming interfaces (API) integration.
- **Web Technologies:** Using Hyper Text Markup Language (HTML), Cascading Style Sheets (CSS), and JavaScript to power the frontend with a responsive, intuitive, and accessible user interface compatible across devices.
- **SQL Database:** Offers lightweight, efficient storage for user data, entrance exam information, and study materials, ensuring quick access and simplified deployment.
- **Amazon Web Services (AWS) Cloud Services:** Delivers a secure hosting environment with built-in security features such as encryption, data protection, and compliance with industry standards, safeguarding sensitive user information and supporting scalable operations.
- **Cybersecurity Measures:** Integrates advanced security protocols, including role-based access control, data encryption, and secure API communication, to protect student information and system integrity.

1.2.2 Key Features

The WAV2T platform is designed with essential features to support students at every stage of their educational journey:

- **User Authentication:** Implements a secure, role-based login system, providing access to tailored content for prospective, current, and graduated students.
- **Entrance Exams:** Includes randomized, timed entrance exams for three distinct pathways—Server and Cloud Application (SCA), Cloud Application Development (CAD), and Cybersecurity Administration (Cyber)—ensuring proper placement and readiness evaluation.
- **Study Materials:** Offers a curated repository of study resources, including videos, written materials, and practice exams. These resources are designed to support entrance preparation, ongoing coursework, and professional certifications.
- **Adaptive Practice Exams:** Leverages a machine learning model to dynamically tailor exam difficulty based on the user's performance and learning patterns. By analyzing factors such as accuracy, response time, and progression, the system provides personalized question sets and targeted feedback, optimizing learning outcomes and engagement.
- **Chatbot Study Buddy:** Features an AI-powered assistant, integrated via AWS's Amazon Lex, to help users with study-related queries and navigation.
- **Push Notifications:** Sends alerts exclusively to WAV2T personnel, notifying them when a test requires grading. This ensures prompt evaluation and timely feedback for users.
- **Multi-Tabbed Interface:** Facilitates seamless navigation with organized tabs for each educational pathway, ensuring intuitive access to exams, study materials, and resources.

2. System Overview

2.1 Product Perspective

The WAV2T system is a secure educational platform designed to support prospective, current, and graduated students by providing a centralized location for entrance exams, study materials, adaptive practice exams, and certification preparation. The system

employs a machine learning model to deliver personalized assessments, enhancing user engagement and optimizing learning outcomes.

The platform maximizes user engagement by offering structured, adaptive study materials and ensuring the secure handling of sensitive user data. Students are guided through the educational process with targeted resources and assessments tailored to their respective pathways (SCA, CAD, Cyber), with support from the integrated chatbot.

This platform addresses the need for a seamless and secure integration of exam administration, personalized study resources, and efficient user data management. It incorporates push notification functionality to alert WAV2T personnel when tests require grading, ensuring timely feedback. By adhering to the latest cybersecurity protocols, the WAV2T system delivers an intuitive, tailored experience while safeguarding personal and academic information.

2.1.1 Relationship to Existing Systems

WAV2T will be developed as an independent platform utilizing APIs and standardized data formats such as JavaScript Object Notation (JSON) and Extensible Markup Language (XML) to enable seamless future integration with other academic or student management systems. This approach ensures compatibility with external educational tools and facilitates the secure exchange of data, including exam results, user information, study material usage analytics, and adaptive learning patterns generated by the system's machine learning model. By leveraging AWS cloud infrastructure, WAV2T ensures scalability, comprehensive security, and efficient data management for current and future needs.

2.1.2 Major Components and Subsystem Interconnections

The WAV2T system is built with a layered architecture comprising the following components:

1. User Interface (UI) Layer:

- **Description:**
 - The UI layer provides components for interacting with the platform, allowing students and administrators to navigate the entrance exams, study materials, and account management sections.
- **Components:**
 - **Student Dashboard:** Interface for users to access study materials, view exam results, and manage personal information.

- **Admin Dashboard:** Interface for administrators to manage user accounts, entrance exam questions, and monitor the system.
- **Pathway-Specific Pages:** Individual pages for each educational pathway (SCA, CAD, Cyber) with associated resources, study materials, and exams.
- **User Profile Management:** Interface allowing users to update personal information, such as recovery emails, and change passwords.
- **Technologies:**
 - **ASP.NET MVC:** Provides web applications, Razor views, HTML, CSS, JavaScript, and web accessibility features using accessible rich internet applications (ARIA).

2. Security Layer:

- **Description:**
 - The security layer manages authentication, authorization, and data protection.
- **Components:**
 - **Authentication:** Secure login and logout functionalities.
 - **Authorization:** Role-based access control for administrators, prospective students, current students, and graduates.
 - **Secure Communication:** Implementation of HTTPS to secure data transmissions, secured with AWS Transport Layer Security (TLS) protocols.
 - **Data Encryption:** Encryption of sensitive data, such as passwords and exam results, both in transit and at rest.
 - **AWS Security Features:**
 - **AWS Identity and Access Management (IAM):** Fine-grained permissions for managing access to AWS resources.
 - **AWS Key Management Service (KMS):** Encryption of sensitive data using managed encryption keys.
 - **AWS Web Application Firewall (WAF):** Protections against common web exploits and attacks such as SQL injection or cross-site scripting (XSS).

- **AWS Shield:** Distributed Denial of Service (DDoS) protection to ensure system availability.
- **CloudTrail and CloudWatch Logs:** Enabling audit, monitoring, and anomaly detection for security events.
- **Technologies:**
 - **ASP.NET Core Identity:** Application-level authentication and authorization, encryption libraries, and HTTPS protocols.
 - **AWS TLS Protocols:** Securing data transmission.
 - **AWS IAM, WAF, Shield, and KMS:** Infrastructure-level security and data protection.

3. Presentation Layer:

- **Description:**
 - This layer handles user requests, processes input and presents data to the user.
- **Components:**
 - **Controllers:** Process user requests, handle logic for routing, and invoke service methods to manage business operations.
 - **Views:** Render dynamic content and display data, enabling users to interact with their dashboards, exams, and study materials.
 - **View Models:** Transfer data between controllers and views, ensuring consistency and separation of concerns.
- **Technologies:**
 - **ASP.NET MVC Framework:** Provides a structured model-view-controller design for organizing the web application.
 - **Razor Views:** Dynamically render HTML content using C# for server-side logic.
 - **HTML, CSS, JavaScript:** Create a responsive and accessible user interface, ensuring compatibility across devices and browsers.

- **Web Accessibility Features (ARIA):** Enhance usability for individuals with disabilities by providing semantic roles and properties to dynamic web elements.
- **Bootstrap:** Streamline responsive design and ensure consistency across UI components.

4. Application Layer:

- **Description:**

- This layer contains the core business logic, handling interactions between the presentation and domain layers.

- **Components:**

- **Exam Management Service:** Handles logic for creating, managing, grading, and submitting entrance exams and practice exams. Integrates machine learning for adaptive practice exams, dynamically adjusting question difficulty based on user performance.
- **User Management Service:** Manages registration, authentication, and role-based access controls. Supports operations such as password recovery, user profile updates, and role assignment.
- **Notification System:** Manages push notifications to WAV2T personnel, alerting them when exams require grading. As well as users to their user account when exams have been graded or new study material becomes available.
- **Study Material Service:** Provides access to curated study materials, including videos, documents, and other resources tailored to each educational pathway (SCA, CAD, Cyber). Tracks resource usage analytics for potential insights.
- **Pathway Management:** Guides users through their specific pathway, ensuring they access the relevant resources and exams based on their chosen track.

- **Technologies:**

- **C#:** Implements core business logic.
- **Machine Learning Integration:** Employs a custom-trained model to manage adaptive learning in practice exams.

- **Dependency Injection (DI):** Manages service lifetimes and dependencies efficiently for modular design.

5. Domain Layer:

- **Description:**
 - The domain model represents the core entities and business rules for WAV2T.
- **Components:**
 - **User:** Represents users (students, administrators) with attributes like name, ID numbers, pathway, and role.
 - **Exam:** Represents entrance exams with attributes like questions, time limits, and scores.
 - **Adaptive Logic Practice Exam:** Represents dynamically tailored exams with attributes like question difficulty, user performance metrics, and skill progression tracking, encapsulating the business rules for machine learning-driven adaptive practice exams.
 - **Study Materials:** Represents the resources available to students, including videos, texts, and practice exams.
- **Technologies:**
 - **Entity Framework Core:** Mapping domain entities to SQL database tables.
 - **SQL Database:** Lightweight efficient storage of domain data.
 - **AWS Hosting Integration:** Ensuring the domain layer can scale securely and efficiently within the AWS environment.
 - **Plain Old CLR Objects (POCOs):** For domain entities and to support object-oriented design principles.

6. Persistence Layer:

- **Description:**
 - This layer handles data storage and retrieval from the database.
- **Components:**

- **Entity Framework (EF):** Manages object-relational mapping between domain entities and database tables.
- **Data Access Objects (DAOs):** Encapsulate data access logic for user information, exams, and study materials.
- **Database Connection Management:** Manages efficient and secure interactions with the SQL database.
- **Technologies:**
 - **SQL Database:** A lightweight, file-based database optimized for local and cloud environments, suitable for the WAV2T platform's needs.
 - **Entity Framework Core:** Managing database interactions and mapping domain entities to database tables.
 - **LINQ (Language Integrated Query):** Querying the database in a strongly typed manner within the application.
 - **AWS Integration:** Includes backup and recovery options, as well as potential hosting or storage enhancements using AWS for static files or Amazon RDS for scalability (if the database evolves in the future).

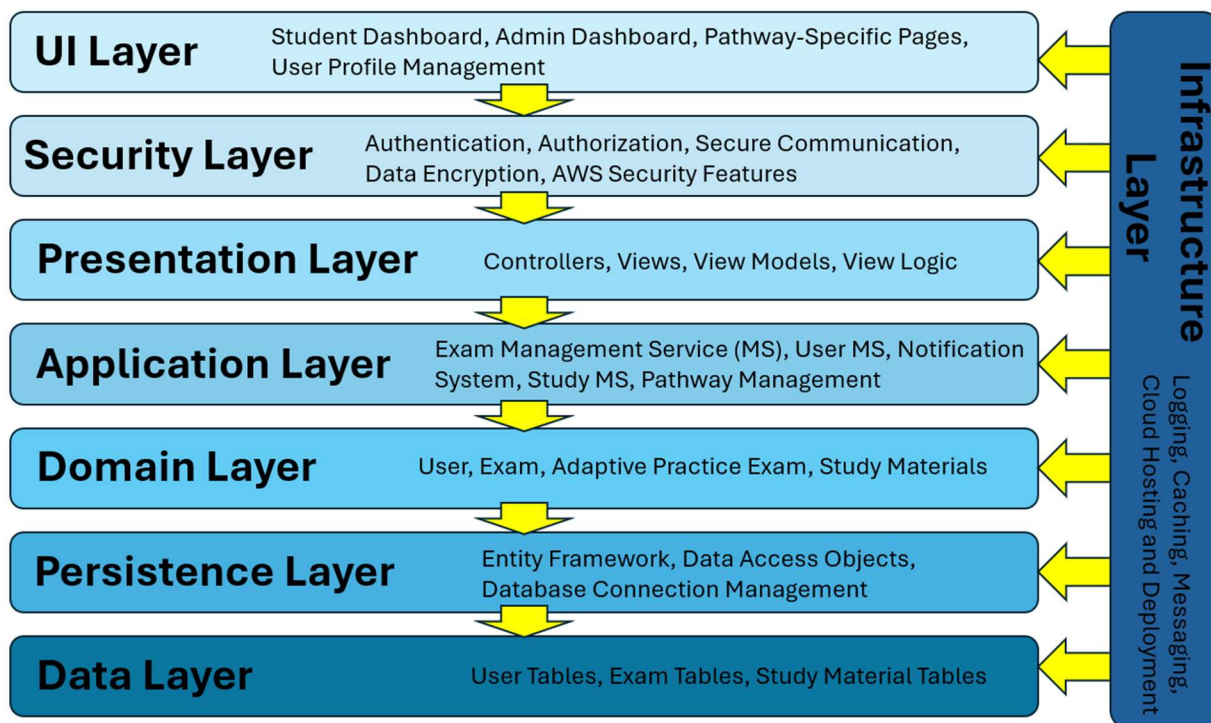
7. Data Layer:

- **Description:**
 - The data layer represents the actual database where information about users, exams, and study materials is stored.
- **Components:**
 - **User Tables:** Store information about users, including personal details, roles, and pathway selections.
 - **Exam Tables:** Store entrance exam questions, answers, results, and performance logs for adaptive learning.
 - **Study Material Tables:** Contain metadata and links for study resources like videos, texts, and practice exams.
- **Technologies:**
 - **SQL Database:** A lightweight, file-based database chosen for its simplicity and efficiency in small to medium-scale applications.

- **Entity Framework Core:** Provides Object-Relational Mapping (ORM) capabilities to map domain entities to database tables and handle CRUD (Create, Read, Update, Delete) operations.
- **AWS Integration:** Data backup and recovery solutions using AWS services.
- **Encryption:** Sensitive data (e.g., user passwords and exam results) is encrypted at rest to ensure security.

8. Infrastructure Layer:

- **Description:**
 - This layer provides supporting services and utilities for the system.
- **Components:**
 - **Logging:** Captures logs for monitoring, security audits, and debugging.
 - **Caching:** Stores frequently accessed data (e.g., study materials and user session data) to enhance performance and reduce database load.
 - **Messaging:** Implements push notifications for WAV2T personnel to alert them about pending exam grading.
 - **Cloud Hosting and Deployment:** Utilizes AWS services for hosting the application, ensuring scalability and high availability.
- **Technologies:**
 - **Logging Frameworks:** Will employ either NLog or Serilog to collect and manage logs.
 - **Caching Library:** Implement Distributed Cache to store session or frequently used data.
 - **Messaging Services:** Use AWS Simple Notification Service (SNS) for notifications.
 - **AWS Cloud Hosting:** Amazon EC2 for hosting the web application, S3 for storing static resources such as videos, or images, and CloudWatch for monitoring application performance and logging.
 - **TLS Protocols:** Ensure secure communication over HTTPS.



2.1.3 External Interfaces

1. **Student Management Systems:** WAV2T platform is designed to integrate with external student management or academic systems using APIs and standardized data formats (e.g., JSON and XML). This enables the synchronization of user data, such as enrollment status, exam scores, and user roles (e.g., prospective, current, or graduated students). These integrations facilitate the seamless import and export of student information, ensuring the WAV2T system remains accurate and up to date.
2. **Email Systems:** System connects to email servers to deliver critical notifications, such as exam grading alerts for administrators, entrance exam results, and updates to study materials. This ensures timely and reliable communication with personnel.
3. **Authentication Services:** Interfaces with external authentication provider, AWS Cognito, to manage secure user authentication. This service supports advanced security measures like single sign-on (SSO) and two-factor authentication (2FA), enhancing the protection of sensitive user data while maintaining ease of access.

2.2 Product Features

The WAV2T system is designed to provide educational support through a range of features that cater to prospective, current, and graduated students. Below is a high-level overview of the key functionalities:

2.2.1 Product Overview

1. User Account Management

- **User Profiles:** Allows users to create and manage their profiles, specifying details like their name, chosen pathway, semester of enrollment, current certifications, and ID numbers (student or application ID). Profiles also allow users to update their recovery email and personal preferences for a personalized experience.
- **Role-Based Access Control:** Implements role-based access to distinguish between prospective students, current students, graduates, and administrators. Each role is granted access only to the resources and features applicable to them.
- **Authentication and Security:** Provides secure login mechanisms, including password protection, encrypted data storage, and options for two-factor authentication (2FA) to protect user accounts and sensitive data. The system leverages AWS security protocols for additional protection.

2. Entrance Exam Management

- **Exam Creation and Administration:** Allows administrators to create and manage three distinct entrance exams, tailored to each educational pathway (Server and Cloud Applications, Cloud Application Development, and Cybersecurity Administration). The exams pull questions from a bank of 500+ items.
- **Exam Randomization and Timing:** Ensures that each student receives a randomized set of 65 questions, consisting of multiple-choice and short-answer formats, and is given 90 minutes to complete the exam.
- **Results and Feedback:** Provides administrators with tools to review submitted exams and publish the results, while notifying students push notification to their account once their results are available.

3. Study Materials and Resources

- **Study Material Repository:** Offers students access to a wide range of study resources, including videos, texts, and practice exams, organized by pathway. The system provides materials to help students prepare for both entrance exams and certifications (e.g., CompTIA Network+, Security+).

- **Adaptive Practice Exams:** Features machine learning-powered adaptive practice exams that adjust question difficulty based on user performance. Students can choose between timed and untimed options, with instant feedback to enhance learning outcomes.
- **Study Tools:** Includes interactive and engaging study tools to assist students in mastering content for entrance exams and certifications.

4. Notification and Reminder System

- **Push Notifications and Email Alerts:** Sends notifications to Vets 2 Tech personnel when exams require grading. Additionally, students are notified about key updates, such as study material releases and exam results.
- **Progress Tracking:** Provides reminders and tracking tools to help users monitor their progress and stay aware of important deadlines.

5. Reporting and Analytics

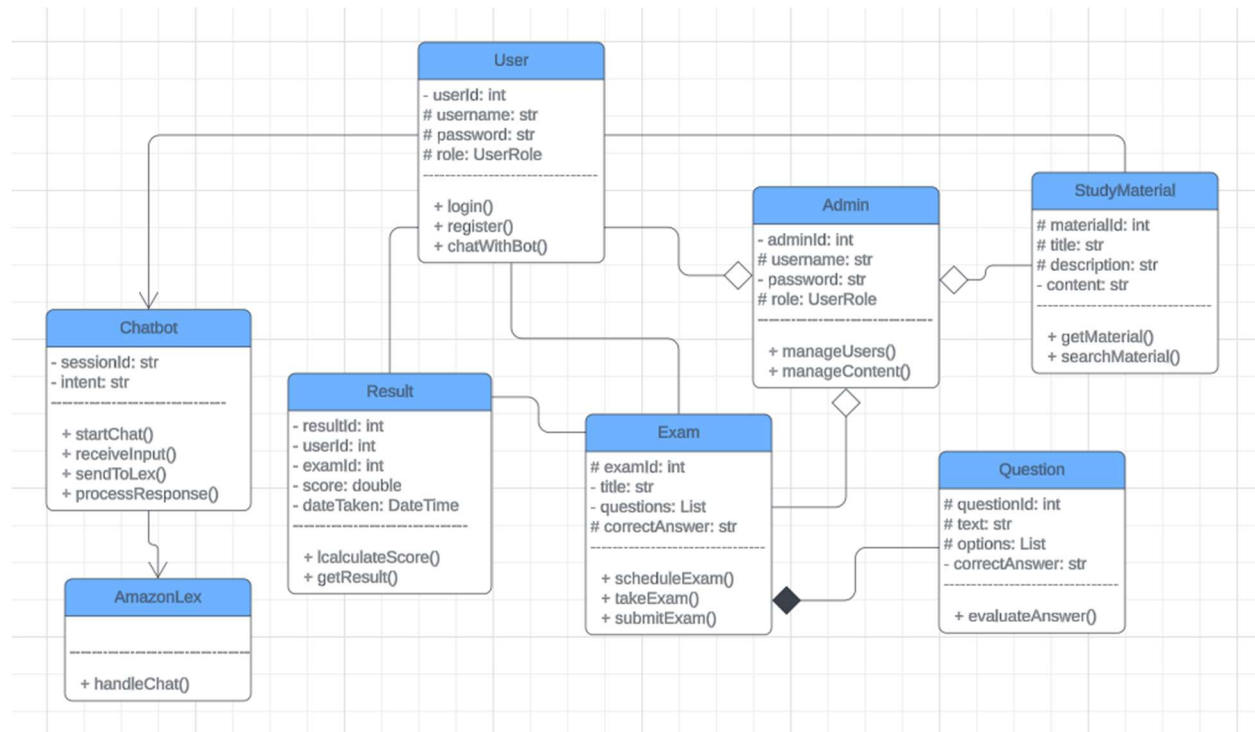
- **Performance Dashboards:** Displays metrics such as exam completion rates, average scores, and user engagement with study materials. Dashboards are available for both administrators and students to track performance.
- **Custom Reports:** Administrators can generate detailed reports on user performance and engagement. Filters by date range, pathway, and user group support data-driven decision-making.
- **Real-Time Analytics:** Provides live insights into user activity and system usage, helping administrators optimize the system and assess the effectiveness of exams and resources.

6. Integration Capabilities

- **API Integration:** Facilitates integration with external systems via APIs using standardized formats such as JSON and XML, allowing the synchronization of user data, exam results, and resource access with institutional systems.
- **Data Import and Export:** Enables the import of user and exam data from external sources and the export of reports and analytics for institutional use or further analysis.

2.2.2 System Interaction Diagram

To visually represent how these features interact within the WAV2T system, a top-level system interaction diagram is provided below. This diagram illustrates the modular structure of the system, showing the relationships between the key components, including the user interface, exam management, study materials, notifications, and the underlying database architecture.



2.3 User Classes and Characteristics

The WAV2T educational platform is designed to support various user classes, each with distinct roles, responsibilities, and interactions with the system. These user groups are defined by their purpose within the educational process, frequency of use, security access levels, and the system features they utilize. Below is a detailed description of the main user classes expected to interact with the WAV2T system:

1. Administrators (Admins)

- **Characteristics:**
 - Staff members or educators responsible for managing the overall system, including user accounts, entrance exams, and study materials.
 - High technical expertise, with a comprehensive understanding of the platform's functionalities and security protocols.

- Frequent users of the system, primarily focusing on exam management, user data, study materials, and monitoring system activity.
- **Privileges:**
 - Full access to all system modules, including account management, exam creation, study material management, and analytics dashboards.
 - Ability to manage user roles, configure system settings, and oversee chatbot interactions and integrations with Amazon Lex.
 - Can configure system settings, manage user roles, and maintain integration with external systems.
 - Notifications to manage exam grading and platform maintenance tasks.
- **Requirements:**
 - Advanced tools for creating and managing exams, study materials, monitoring system logs, and sending targeted notifications.
 - Access to detailed audit logs for tracking platform activity and ensuring compliance with security protocols.

2. Prospective/Incoming Students

- **Characteristics:**
 - Students applying to the institution, primarily interacting with the system to take entrance exams and access preparatory study materials.
 - Low to moderate technical expertise, requiring simple, intuitive interfaces for navigation.
 - Frequent users during the application process, with an emphasis on exam preparation and completion.
- **Privileges:**
 - Access to pathway-specific entrance exams (SCA, CAD, Cyber) and related preparatory study materials.
 - Ability to view exam instructions, complete exams, and receive exam results.
- **Requirements:**

- Clear navigation to exams and study materials, with guidance on using the platform.
- Secure exam environments to maintain academic integrity and user trust.
- Notifications for exam-related updates, including results and deadlines.
- Secure exam environments to maintain academic integrity.

3. Current Students

- **Characteristics:**

- Enrolled students using the platform to access study materials, take practice exams, and manage academic progress.
- Moderate technical expertise with regular platform interactions for learning and receiving system notifications.
- Frequent users of study resources and practice exams, particularly for certifications.

- **Privileges:**

- Access to pathway-specific study materials (e.g., videos, texts, practice exams) for entrance exams and certifications (e.g., CompTIA Security+).
- Ability to track progress, receive feedback on practice exams, and manage learning pathways.
- Notifications for certification deadlines, study material updates, and reminders.

- **Requirements:**

- A well-organized repository of study resources and adaptive practice exams tailored to individual performance.
- Notifications to assist with study planning and deadlines.
- Practice exams with customizable settings for feedback and time limits.
- Cross-platform accessibility for ease of use on multiple devices.

4. Graduated Students

- **Characteristics:**

- Former students accessing the platform for professional development and certification preparation.
- Varied technical expertise, with occasional use for study materials and certification practice exams.
- Infrequent interaction, focused on certification preparation and professional growth.
- **Privileges:**
 - Access to all certification study materials, including adaptive practice exams and professional development resources.
 - Ability to track preparation progress and receive feedback on practice exams.
- **Requirements:**
 - Simplified access to certification resources and tools to help with ongoing professional development.
 - Cross-platform support to allow access from multiple devices, making it easy to study on the go.

5. Support Staff

- **Characteristics:**
 - IT staff or platform administrators responsible for ensuring system performance, security, and troubleshooting.
 - High technical expertise, with deep knowledge of the platform's backend and database architecture.
 - Frequent interaction for maintaining system operations, resolving issues, and implementing updates.
- **Privileges:**
 - Full access to system configuration, database management, and monitoring tools.
 - Ability to troubleshoot technical issues, manage chatbot interactions with Amazon Lex, and ensure compliance with security protocols.
- **Requirements:**